

# Critical Care 雙週期刊回顧

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2026-04-08 ~ 2026-05-05 ( 4 週補回顧 )

謝慕揚 MD, PhD, FESC | 2026-05-05

涵蓋：ICM (Intensive Care Medicine)、Critical Care、CCM (Critical Care Medicine)、Lancet Respir Med

常用縮寫對照 ( 一 )

| 縮寫                                                      | 全名                                                                                                         |
|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| RCT                                                     | Randomized Controlled Trial 隨機對照試驗                                                                         |
| OR / aOR                                                | (Adjusted) Odds Ratio 校正勝算比                                                                                |
| HR                                                      | Hazard Ratio 風險比                                                                                           |
| CI / CrI                                                | Confidence / Credible Interval 信賴區間                                                                        |
| IPD-MA                                                  | Individual Participant Data Meta-Analysis                                                                  |
| NMA                                                     | Network Meta-Analysis 網絡統合分析                                                                               |
| ICU / PICU                                              | Intensive Care Unit / Pediatric ICU                                                                        |
| ED / NHS                                                | Emergency Department / National Health Service                                                             |
| ARDS / AHRF / ARF                                       | Acute Resp Distress Synd / Acute Hypoxaemic / Acute Resp Failure                                           |
| NIV / IMV                                               | Non-Invasive / Invasive Mechanical Ventilation                                                             |
| CPAP / HFNC / NIPPV                                     | Continuous Positive Airway Pressure / High-Flow Nasal Cannula / Non-Invasive Positive Pressure Ventilation |
| SOT                                                     | Standard Oxygen Therapy ( 呼吸文獻 ) / Solid Organ Transplant ( 移植 )                                           |
| PaCO <sub>2</sub> / PaO <sub>2</sub> / FiO <sub>2</sub> | Arterial CO <sub>2</sub> / O <sub>2</sub> partial pressure / Fraction of inspired O <sub>2</sub>           |

常用縮寫對照 ( 二 )

| 縮寫                                          | 全名                                                                                |
|---------------------------------------------|-----------------------------------------------------------------------------------|
| AKI / AKD / CKD                             | Acute Kidney Injury / Disease, Chronic Kidney Disease                             |
| RRT / CKRT / UF                             | Renal Replacement Therapy / Continuous Kidney Replacement / Ultrafiltration       |
| KDIGO                                       | Kidney Disease: Improving Global Outcomes                                         |
| MP / VILI                                   | Mechanical Power / Ventilator-Induced Lung Injury                                 |
| ECCO <sub>2</sub> R / miECCO <sub>2</sub> R | (Minimally Invasive) Extracorporeal CO <sub>2</sub> Removal                       |
| TBI / ICP / ICH                             | Traumatic Brain Injury / Intracranial Pressure / Hypertension                     |
| TCD / ONSD / EEG / MRI                      | Transcranial Doppler / Optic Nerve Sheath Diameter / EEG / MRI                    |
| PCT / NEWS2                                 | Procalcitonin / National Early Warning Score 2                                    |
| SOFA / APACHE-II                            | Sequential Organ Failure Assessment / Acute Physiology And Chronic Health Eval II |
| EAA / PMX                                   | Endotoxin Activity Assay / Polymyxin B haemoadsorption                            |
| SSC / SOC                                   | Surviving Sepsis Campaign / Standard of Care                                      |
| PE / PA                                     | Pulmonary Embolism / Pulmonary Artery                                             |
| FDA / PMA                                   | Food and Drug Administration / Premarket Approval                                 |
| WGS / RR-TB / AI                            | Whole Genome Sequencing / Rifampicin-Resistant TB / Artificial Intelligence       |
| GMR / AUC / aRD                             | Geometric Mean Ratio / Area Under Curve / Adjusted Risk Difference                |

# 重點摘要 (Top Pearls)

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1. **Tigris** — Polymyxin B (PMX) 在 endotoxic septic shock 後驗益處 95.3% ( 28d ) / 99.4% ( 90d ) — 但僅 157 人、open-label
2. **PRONTO** — Procalcitonin (PCT) + NEWS2 主要終點 negative，mortality 卻 -3.12% ( 機制未明 )
3. **AHRF NMA** — CPAP ( OR 0.45 ) / HFNC ( OR 0.61 ) 皆減插管；CPAP 也減死亡
4. **Thille NIV** — 高風險、無 hypercapnia 病人，NIV+HFNC 7d 再插管 11.8% vs 17.6%
5. **MP score** — 拋棄單一閾值；compliance + duration 共同決定 VILI ( ventilator-induced lung injury ) 風險

## 重點摘要（續）

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- 6. **AKI biomarker bundle** — KDIGO bundle + biomarker enrichment : AKI 17.7% vs 27.1% ( OR 0.55 )
- 7. **NUTRIREA-3 post hoc** — 低熱量低蛋白餵食對腎臟安全
- 8. **Brain temperature** — 從強制低溫轉向「fever 偵測 + normothermia」
- 9. **BESS** — 嬰兒 bronchiolitis 用 surfactant **無效**
- 10. **Immunocompromised ARF** — 30 天死亡 47.3% ; HFNC 為保護因子

# 1. Sepsis 與血液淨化

# Tigris : PMX 在 endotoxic septic shock — Phase 3 RCT

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Neyra JA, et al. Lancet Respir Med 2026;14(5) | PMID 41887242

- **設計** : 19 個美國醫院、open-label、Bayesian phase 3、2:1 randomization
- **族群** : 成人 septic shock + multiorgan failure + **Endotoxin Activity Assay (EAA)**  
**0.60-0.89**
- **介入** : PMX (Polymyxin B haemoadsorption) 兩次 (90-120 min, 22h apart) vs Standard of Care (SOC)
- **N** : 157 ( 106 / 51 ) — 篩選 14,890 人才入選
- **主要終點** : 28-day mortality

# Tigris : 結果

Neyra JA, et al. Lancet Respir Med 2026

| 終點                         | PMX | Control | OR (95% CrI)     | Posterior P(benefit) |
|----------------------------|-----|---------|------------------|----------------------|
| 28-day death               | 39% | 45%     | 0.67 (0.39-1.08) | 95.3%                |
| 90-day death               | —   | —       | 0.54 (0.32-0.87) | 99.4%                |
| Serious AE (Adverse Event) | 30% | 22%     | diff -8%         | —                    |

**解讀重點**：Bayesian 先驗來自 EUPHRATES 次族群分析；族群高度 enriched ( EAA 0.60-0.89 )  
— 不可外推到無 EAA 測量的一般 sepsis



# PRONTO : PCT + NEWS2 在 ED Sepsis

Todd S, et al. Lancet Respir Med 2026;14(5) | PMID 41881047

- 英國 NHS ( National Health Service ) 17 Trust 、 20 Emergency Department (ED)
- **n=7,667** randomised
- 比較 : PCT (Procalcitonin) + NEWS2 (National Early Warning Score 2) algorithm vs 標準 NEWS2
- **共同主要終點** : 3h IV (intravenous) ABX (antibiotics) ( superiority ) + 28d mortality ( NI margin 2.5% )

| 終點        | PCT 組 | 標準組   | 結果                  |
|-----------|-------|-------|---------------------|
| 3h IV ABX | 48.4% | 48.2% | 無差異 (p=0.95)        |
| 28d 死亡    | 13.6% | 16.6% | aRD -3.12% p=0.0009 |

## 2. Mechanical Ventilation

# AHRF 非侵襲呼吸支持 NMA

Lee KG, et al. Lancet Respir Med 2026 | PMID 42034114

44 RCT , 9,704 人 ; vs Standard Oxygen Therapy (SOT) :

| 介入                                                  | Intubation OR      | Mortality OR       | GRADE     |
|-----------------------------------------------------|--------------------|--------------------|-----------|
| CPAP (Continuous Positive Airway Pressure)          | 0.45 (0.27-0.72) ↓ | 0.73 (0.55-0.95) ↓ | Mod / Low |
| HFNC (High-Flow Nasal Cannula)                      | 0.61 (0.42-0.86) ↓ | 0.83 (0.66-0.98) ↓ | Mod / Low |
| Bilevel NIPPV (Non-Invasive Positive Pressure Vent) | 0.60 (0.39-0.89) ↓ | 0.93 (0.71-1.17) — | Mod / Low |

CPAP 在減插管最強 ; CPAP 與 HFNC 為 mortality 兩個最有依據選項

# Extubation 後預防性 NIV — 非 hypercapnia 也有用

Thille AW, et al. ICM 2026 | PMID 41973108

- 兩 RCT post hoc ; n=829 ( 高風險、 $\text{PaCO}_2 \leq 45 \text{ mmHg}$  )
- **NIV (Non-Invasive Ventilation) + HFNC vs HFNC alone**

| 終點     | NIV+HFNC | HFNC only | $\Delta$        |
|--------|----------|-----------|-----------------|
| 7d 再插管 | 11.8%    | 17.6%     | -5.8% (p=0.021) |

高齡 (>65) 或心肺合併症病人即使無 hypercapnia 仍可受益於 NIV

# Risk-Adjusted Mechanical Power Score

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Lijović L, et al. Crit Care Med 2026 | PMID 41945715

- n=2,150 中-重度 ARDS ( Acute Respiratory Distress Syndrome ; NL + US 兩大資料集 )
- **High-compliance lung** : dose-response , 10 J/min 起 hazard ↑ , **累積有害**
- **Low-compliance lung** : 風險集中 11-20 J/min 窄帶、**無累積效應**
- AUC (Area Under Curve) **0.863** ( XGBoost / eXtreme Gradient Boosting )

「單一 MP (Mechanical Power) 閾值」已不足夠 — 需 power × duration × compliance 三維度評估

# 3. AKI / Renal / Nutrition

# 高風險術後 AKI : KDIGO Bundle + Biomarker

von Groote T, et al. ICM 2026 | PMID 42007986

- 4 RCT IPD-MA ( Individual Participant Data Meta-Analysis ) ; **n=1,851**
- 介入 : KDIGO (Kidney Disease: Improving Global Outcomes) bundle + biomarker enrichment

| 終點                            | Intervention | Control | OR (95% CI)      |
|-------------------------------|--------------|---------|------------------|
| 中-重度 AKI (KDIGO $\geq$ 2) 72h | 17.7%        | 27.1%   | 0.55 (0.44-0.70) |

$I^2 = 0\%$  ( 完全一致 ) ;  $p < 0.0001$

高風險術後病人若有 enrichment biomarker , KDIGO bundle 應為標準

# NUTRIREA-3 Post Hoc：低熱量低蛋白不傷腎

Schleef M, et al. ICM 2026 | PMID 41989569

- 3,036 機械通氣 + shock；前 7 天
- 低 ( 6 kcal/kg + 0.2-0.4g protein/kg/d ) vs 標準 ( 25 kcal/kg + 1.0-1.3g/kg/d )

| 終點                                    | 低熱量   | 標準    | HR (Hazard Ratio, 95% CI) |
|---------------------------------------|-------|-------|---------------------------|
| AKD (Acute Kidney Disease) during ICU | 44.6% | 46.1% | 0.97 (0.88-1.07)          |
| Peak urea                             | ↓     | ↑     | p=0.002                   |

休克急性期低熱量低蛋白餵食策略對腎臟安全



# 4. 神經重症

# 急性腦損傷溫度管理：觀念演進

Lavinio A, et al. ICM 2026 | PMID 42007984

| 情境                                  | 過去      | 現在                         |
|-------------------------------------|---------|----------------------------|
| <b>TBI</b> (Traumatic Brain Injury) | 預防性低溫   | 僅 refractory ICH 才用        |
| <b>Acute vascular brain injury</b>  | 強制低溫    | 早期 fever 偵測 + normothermia |
| <b>Post-cardiac arrest</b>          | 強制 33°C | 32-37.5°C 任選 + 預防 fever    |

神經 ICU 溫度管理 = **預防發燒 + maintained normothermia**

ICH = Intracranial Hypertension ( 顱內高壓 )

# Non-Invasive ICP ( Intracranial Pressure ) 工具

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Picetti E, et al. ICM 2026 | PMID 42043555

- **工具** :
  - TCD (Transcranial Doppler , 經顱都卜勒)
  - ONSD (Optic Nerve Sheath Diameter , 視神經鞘直徑)
  - Pupillometry ( 自動瞳孔測量 ) 、 skull compliance 、 time-of-flight US
- **限制** : calibration / zeroing — 絕對準確度受限
  - TCD  $\pm 7-15$  mmHg ; ONSD  $\pm 7-10$  mmHg
- **適用** : 選 invasive monitor 時機 ; cardiac arrest 、 liver failure 、 sepsis 後評估 ; **低資源地區**

# 5. ICU 復健與其他

# 重症復健標準化建議

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Hodgson CL, et al. ICM 2026 | PMID 42059920

- **核心**：早期 awakening + mobilisation、安全 ( low AE rate )
- **三面向**：身體、認知、心理 — 跨 ICU、病房、社區
- **未來方向**：詳細記錄 dose ( time、intensity、duration )、實作策略

將復健視為與抗生素、ventilation 同等的核心治療，從入院 day 1 啟動

# Honorable Mentions

| 試驗 / 主題                                                                                                               | 結論                                                   | 連結                            |
|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|-------------------------------|
| <b>BESS</b> infant bronchiolitis surfactant                                                                           | <b>NEGATIVE</b> (n=232; IMV 64.9 vs 62.0h, GMR 1.02) | <a href="#">PMID 41875912</a> |
| Intermediate-risk PE (Pulmonary Embolism) thrombectomy timing                                                         | mortality 無差，PA 壓力較低                                 | <a href="#">PMID 42023946</a> |
| Immunocompromised ARF (n=9,854)                                                                                       | 30d mortality <b>47.3%</b> ；HFNC 保護                  | <a href="#">PMID 41856149</a> |
| <b>SOFA-2</b> (Sequential Organ Failure Assessment 2)                                                                 | 優於 SOFA-1 ( 兩 cohort )                               | <a href="#">PMID 42036712</a> |
| <b>SMARTT</b> WGS (Whole Genome Sequencing)-guided RR-TB                                                              | unfavourable outcomes ↓                              | <a href="#">PMID 42026006</a> |
| miECCO <sub>2</sub> R (minimally invasive Extracorporeal CO <sub>2</sub> Removal) via RRT (Renal Replacement Therapy) | feasible，PaCO <sub>2</sub> 71→52 in 24h              | <a href="#">PMID 42057044</a> |

# Take Home Messages

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1. **Sepsis** : Tigris 顯示 PMX 在高 EAA 族群可能有效，但族群極窄；PRONTO 提醒 PCT 不能取代臨床判斷
2. **Ventilation** : CPAP 與 HFNC 為 AHRF 兩大首選；MP 風險需考量 compliance；高風險 extubation 用 NIV+HFNC
3. **AKI** : KDIGO bundle + biomarker enrichment 可減半中-重度 AKI；低熱量低蛋白不傷腎
4. **Neurocritical** : normothermia 取代強制低溫；non-invasive ICP 為輔助工具
5. **Whole-ICU mindset** : 復健為核心治療而非附加項目

## 主要參考文獻

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1. Neyra JA, et al. Tigris. *Lancet Respir Med* 2026;14(5) PMID 41887242
2. Todd S, et al. PRONTO. *Lancet Respir Med* 2026;14(5) PMID 41881047
3. Lee KG, et al. AHRF NMA. *Lancet Respir Med* 2026 PMID 42034114
4. Thille AW, et al. NIV after extubation. *ICM* 2026 PMID 41973108
5. Lijović L, et al. MP score. *Crit Care Med* 2026 PMID 41945715
6. Kryvenko V, et al. miECCO<sub>2</sub>R. *Crit Care* 2026;30(1) PMID 42057044
7. von Groote T, et al. AKI bundle IPD-MA. *ICM* 2026 PMID 42007986
8. Murugan R, et al. UF rate trial. *Crit Care* 2026;30(1) PMID 42021350
9. Schleef M, et al. NUTRIREA-3 post hoc. *ICM* 2026 PMID 41989569
10. Lavinio A, et al. Brain temperature. *ICM* 2026 PMID 42007984



### 主要參考文獻 ( 續 )

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11. Picetti E, et al. Non-invasive ICP. *ICM 2026* PMID 42043555
12. Silva S, et al. Coma stepwise. *ICM 2026* PMID 42059919
13. Hodgson CL, et al. ICU rehabilitation. *ICM 2026* PMID 42059920
14. Semple MG, et al. BESS. *Lancet Respir Med* 2026;14(5) PMID 41875912
15. Chiang CJ, et al. PE thrombectomy timing. *Crit Care Med* 2026 PMID 42023946
16. Azoulay E, et al. Immunocompromised ARF. *Lancet Respir Med* 2026;14(5) PMID 41856149
17. Liufu R, et al. SOFA-2. *Crit Care* 2026 PMID 42036712
18. Wei S, et al. SOFA-2 pneumonia. *Crit Care* 2026 PMID 42045961
19. Van Rie A, et al. SMARTT. *Lancet Respir Med* 2026 PMID 42026006
20. Mallmann C, et al. Vasopressor weaning. *ICM 2026* PMID 41973110

# 謝謝聆聽

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Q & A

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下次回顧：2026-05-15 (排程已建立)